

(a)

Raw SIM stack

SIM stack 1

SIM stack 2

Noisy SR
image 2

Target

*Loss function*Denoiser loss
L2 term and SSIM term

Output 1

Decoder loss
L2 term and TV term *Resampling* *SIM
Reconstruction* *Convolution* *LeakyReLU* *Batch norm* *Max Pooling* *Upsample*Noisy SR
image 1

(b)

Stage 1. Denoising

Input

Stage 2. Enhancement

Downsampling

Upsampling

Output 2

Clear SR
image

Downsample block

Upsample block